MILITARY SPECIFICATION

GRENADE, HAND, PRACTICE, M30 W/FUZE M205A2, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of grenade designated as the M30 Practice Hand Grenade.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 —Powder, Black
MIL-A-2550 —Ammunition and Special Weapons; General Specification for
MIL-I-45208 —Inspection Requirements, General Specification for
MIL-I-45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105D —Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 —Quality Assurance Terms and Definitions
MIL-STD-1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8861647—Grenade, Hand, Practice, M30, W/Fuze, M205A2 Assembly.
7548339—Container, Fiber, Ammunition for Grenade, Hand, Practice, M30, W/Fuze M205A2 Assembly.
7548340—Box, Packing, Ammunition, for Grenade, Hand, Practice, M30 W/Fuze M205A2, Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-8781229—Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

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3.1 **Material.** Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 **Grenade assembly.** The grenade assembly shall comply with all requirements specified on Drawing (dwg.) 8861647 and with all requirements specified in applicable specifications.

3.3 **Functioning.** The grenade assembly shall emit a cloud of smoke and a loud report, and shall show no evidence of rupture or separation of the metal parts, when tested as specified in 4.3.1.

3.4 **Moisture content of black powder.** The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.5 percent, when tested as specified in 4.3.2.

3.5 **Waterproofness of charge assembly.** The charge assembly shall show no evidence of leakage when tested as specified in 4.3.3.

3.6 **Workmanship.** All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, chips, scale, drit, grease, rust, salt deposits and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** Supplier responsibility will be as specified in the contract or purchase order. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.4. The reports shall contain, as a minimum, the following information:

(a) Contractor and contract number.
(b) Number and revision date of specification.
(c) Characteristic code number.
(d) Lot or serial number.
(e) Number of defectives found.
(f) Assignable cause or analysis of occurrence.
(g) Corrective action taken or recommended.

When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all ma-

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terial procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of grenade assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

(a) Components of one lot interfix number from one manufacturer.

(b) Black powder from not more than one ammunition lot from one manufacturer.

(c) M205A2 practice grenade fuzes of one lot interfix number from one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.2.4.

4.2.2.1 Charge assembly (see dwg. 8861650 covering a detail of dwg. 8861647)

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Weight of charge	Balance	01001
102.	Bag torn or perforated	Visual	01002
103.	Heat seal wrinkled or incomplete	Visual	01003
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	01004

4.2.2.2 Assembly, prior to assembling fuze (see dwg. 8861647).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Charge assembly missing	Visual	02001
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02002

4.2.2.3 Assembly (see dwg. 8861647).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Safety pin not properly spread	Visual	03001*

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Categories	Defects	Method of inspection	Code No.
Major:	AQL 0.25 percent		
101.	Incorrect fuze	Visual	03002
102.	Component missing	Visual	03003
Minor:	AQL 1.50 percent		
201.	Fuze loose	Manual	03004
202.	Protective coating contains bare spot	Visual	03005
203.	Marking misleading or unidentifiable, including incorrect color code	Visual	03006
204.	Evidence of poor workmanship (see 3.6)	Visual	03007

4.2.2.4 Sealed container (see dwg. 7548339)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Tape incompete or badly wrinkled	Visual	04001
102.	Metal end loose or distorted	Visual-Manual	04002
103.	Cut or damage through all layers	Visual-Manual	04003
Minor:	AQL 1.50 percent		
201.	Length of tear tab, minimum (min.)	Scale	04004
202.	Color of tape incorrect	Visual	04005
203.	Marking misleading or unidentifiable	Visual	04006
204.	Contents loose	Manual	04007

4.2.2.5 Sealed packing box (see dwg. 7548340)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Board broken to extent that contents are exposed or liable to become exposed	Visual	05001
102.	DOD symbol misleading or unidentifiable	Visual	05002
103.	Hardware missing, broken, or loose	Visual-Manual	05003
104.	Strapping missing, broken, or loose	Visual-Manual	05004
Minor:	AQL 2.50 percent		
201.	Marking misleading or unidentifiable	Visual	05005
202.	Hardware or strapping improperly engaged	Visual-Manual	05006
203.	Car seal missing, unsealed, or improperly positioned	Visual-Manual	05007
204.	Handle missing or insecure	Visual-Manual	05008
205.	Contents loose	Manual	05009

4.2.3 *Production testing.* The following tests shall be performed using Standard MIL–STD–105D for sampling plan requirements unless otherwise specified.

4.2.3.1 Functioning (see 3.3) Major defect, Code No. 03009.

4.2.3.1.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the specified acceptance criteria, the sampling plan for this test shall be in accordance with Code letter L, and AQL of 0.25 percent. The test shall be performed as specified in 4.3.1.

4.2.3.1.2 After three consecutive lots have complied with the criteria of 4.2.3.1.1, the sampling plan shall be in accordance with Code letter H, and AQL of 0.25 percent.

4.2.3.2 *Moisture content of black powder (see 3.4) Major defect, Code No. 06001.* The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of black powder from each eight hours production of charge assemblies. Composite samples shall not be used. The test shall be performed as specified in 4.3.2.

4.2.3.3 Waterproofness of charge assembly (see 3.5) Major defect, Code No. 01006.

4.2.3.3.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the specified criteria, the sampling plan for this test shall be in accordance with Code letter M, and AQL of 0.040 percent. The test shall be performed as specified in 4.3.3.

4.2.3.3.2 After three consecutive lots have complied with the criteria of 4.2.3.3.1, the

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sampling plan shall be in accordance with Code letter K, and AQL of 0.10 percent.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Number IEL–8781229 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Functioning.* The grenade assembly shall be placed in a holding fixture and fired by removing the safety pin. The grenade assembly shall be observed for compliance with the requirements.

4.3.2 *Moisture content of black powder.* The moisture content of the black powder shall be determined as specified in Specification MIL–P–223.

4.3.3 *Waterproofness of charge assembly.* The charge assembly shall be conditioned at room temperature for 4 hours, then immersed in water heated to a temperature of at least 50 degrees Fahrenheit above the temperature at which the charge assembly was conditioned. The assembly shall be parallel to and not less than one inch below the surface of the water. The assembly shall be rotated until the entire surface has been examined. Each seam and surface of the charge assembly shall be observed for minimum period of 15 seconds, and a total elapsed time not to exceed two minutes. A steady stream or recurring succession of bubbles from any surface or seam shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The grenade assemblies shall be preserved and packaged in accordance with dwg. 7548339.

5.2 Packing.

5.2.1 *Level A.* The fiber containers shall be packed in accordance with dwg. 7548340.

5.3 Marking. Marking shall be as specified on dwgs. 7548339 and 7548340.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

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6.3.1 *Inspection Equipment Lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection Equipment List codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor's design responsibility on contractor format.
OD —Ordnance design.
ODMU—Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification control drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and cali-

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bration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of critical defect reports. For U. S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP–RO; Picatinny Arsenal, ATTN: SMUPA–ND.

6.4.1 *Reporting to other services.* For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

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